3.01	Powers and roots				
3.01a	Index notation	Use positive integer indices to write, for example, $2 \times 2 \times 2 \times 2 = 2^4$	Use negative integer indices to represent reciprocals.	Use fractional indices to represent roots and combinations of powers and roots.	N6, N7
3.01b	Calculation and estimation of powers and roots	Calculate positive integer powers and exact roots. e.g. $2^4 = 16$ $\sqrt{9} = 3$ $\sqrt[3]{8} = 2$ Recognise simple powers of 2, 3, 4 and 5. e.g. $27 = 3^3$ *[see also Inverse operations, 1.04a]*	Calculate with integer powers. e.g. $2^{-3} = \frac{1}{8}$ Calculate with roots.	Calculate fractional powers. e.g. $16^{\frac{-3}{4}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{8}$ Estimate powers and roots. e.g. $\sqrt{51}$ to the nearest whole number	N6, N7
3.01c	Laws of indices	*[see also Simplifying products and quotients, 6.01c]*	Know and apply: $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$ *[see also Calculations with numbers in standard form, 3.02b, Simplifying products and quotients, 6.01c]*		N7, A4

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.